JAN–S–230

26 JUNE 1945

JOINT ARMY-NAVY SPECIFICATION

SILICON, POWDERED

Army Number
57-25B

Navy Number
51S55

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army
57–25A
14 Dec. 1942

Navy Department

A. APPLICABLE SPECIFICATIONS

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–110—Packaging and Packing for Overseas Shipment—Drums: Metal, 55-Gallon (For Other Than Petroleum Products).

JAN–P–118—Packaging and Packing for Overseas Shipment, Sacks, Paper, Shipping, Multiwall.

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms Ammunition.[1]

100–2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing.

INTERSTATE COMMERCE COMMISSION SPECIFICATIONS

37E—Steel Drums.

B. GRADES AND CLASSES

B–1. This specification covers the following grades and classes of silicon as specified in the contract or order. (See par. H–1.)

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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Grade I:
Class A.
Class B.

Grade II:
Class C.

C. MATERIAL AND WORKMANSHIP

C–1. See section E.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Chemical requirements.*—Powdered silicon shall conform to the chemical requirements shown in table I.

TABLE I

	Grade I	Grade II
	Percent	*Percent*
Silicon (minimum)	97.0	96.5
Iron (maximum)	1.00	1.50
Aluminum (maximum)	1.00	1.00

E–2. *Granulation.*—Powdered silicon shall conform to the granulation requirements shown in table II for the class specified using U. S. standard sieves conforming to the requirements of Federal Specification RR–S–366.

TABLE II

	Class A	Class B	Class C
Through 149-micron (No. 100) sieve (minimum)	99.0		
Through 88-micron (No. 170) sieve (minimum)			98
Through 74-micron (No. 200) sieve (minimum)	80.0	99.9	
Through 62-micron (No. 230) sieve (minimum)			90

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lots.*—A lot shall not exceed 2,000 pounds.

F–2. *Sampling.*—A minimum of 10 percent of the containers in the lot shall be selected in such a manner as to be representative of the lot. When lots comprise less than 100 containers either 10 containers or all containers in the lot shall be selected. Using a thief or scoop, remove from different parts of each selected container sufficient material to give a primary sample of approximately 12 ounces. Mix the sample thoroughly and place an 8-ounce portion in a bottle so labeled that the container from which the sample was taken can be easily identified. The remainder of each sample shall be mixed thoroughly and quartered until a composite sample of approximately 8 ounces is obtained. Place the composite sample in a rubber stoppered bottle and label so as to show the name of the material, manufacturer, plant, contract or order number, number of pounds in the lot and lot number. All acceptance tests shall be made on the composite sample representative of the lot. The primary samples shall be held by the Government inspector for possible future examination should the composite sample fail to meet the requirements.

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F–3. *Inspection.*—Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.

F–4. *Tests.*—The following tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Silicon.*—Transfer a weighed portion of approximately 0.500 gm. of the sample to a large platinum crucible. Add to the crucible a mixture of 10 gm. of anhydrous sodium carbonate and 1 gm. of potassium nitrate and mix thoroughly with the sample. Fuse the contents of the crucible carefully, cool, and dissolve the melt in 200 ml. of 22-percent hydrochloric acid solution in a large porcelain casserole or platinum dish. Clean the crucible carefully and add the washings to the main solution. Evaporate the solution carefully to dryness. Add 10 ml. of concentrated hydrochloric acid, warm, add 100 ml. of water, and heat until all soluble salts have dissolved. Filter the solution and wash the residue thoroughly with hot water. Evaporate the combined filtrate and washings to dryness and dissolve, filter, and wash as before. Save the combined filtrate and washings for the determination of iron and aluminum. Dry the residue in a platinum crucible provided with a cover, and ignite carefully to constant weight. Cool the crucible and residue in a sulfuric acid desiccator and weigh with the cover on. Moisten the silica with a few drops of 50-percent sulfuric acid solution and then carefully add hydrofluoric acid until the crucible is approximately three-fourths full. Evaporate the solution to dryness, ignite, and weigh. Run a blank determination on all reagents including the hydrofluoric acid, and make any necessary corrections. Calculate the percentage of silicon in the sample as follows:

$$\text{Percentage of silicon} = \frac{46.72\,A}{W}$$

where

A = loss in weight of crucible.
W = weight of sample.

F–4b. *Iron.*—Fuse the residue remaining in the platinum crucible after the volatilization of the silica as prescribed in paragraph F–4a, with a small amount of potassium pyrosulfate. Dissolve the melt in water and add the solution to the combined filtrate and washings from the second evaporation to dryness as prescribed in paragraph F–4a. Heat the solution to 80° to 90° C. and saturate thoroughly with hydrogen sulfide. Let the solution stand for approximately 30 minutes, filter, and wash with acidified hydrogen sulfide water. Boil until all hydrogen sulfide has been expelled, and oxidize the iron by the addition of bromine water until the solution becomes colored distinctly yellow. Boil until all bromine has been expelled, and adjust the volume of the solution to approximately 150 ml. Make the solution just ammoniacal to methyl red, boil for a few minutes, filter through a filter paper, and wash the precipitate with hot 1-percent ammonium chloride solution. Dissolve the precipitate on the filter paper with hot, 22-percent hydrochloric acid, and wash the paper thoroughly with hot water. Reprecipitate the iron and aluminum, filter, and wash as directed above. Ignite the precipitate in a tared platinum crucible, cool, and weigh. The increase in weight represents the combined iron and aluminum as ferric oxide and aluminum oxide. Represent the weight by the symbol A. Fuse the residue in the

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crucible with a small amount of potassium pyrosulfate. Dissolve the melt in 100 ml. of hot water containing 5 ml. of 50-percent sulfuric acid solution, dilute the solution to approximately 125 ml. and pass in H_2S gas for approximately 30 minutes. Filter the solution and wash the precipitate with hydrogen sulfide water acidified with sulfuric acid. Boil until all hydrogen sulfide is removed from the combined filtrate and washings, cool the solution and titrate with 0.1 normal potassium permanganate solution. Calculate the percentage of iron in the sample as follows:

$$\text{Percentage of iron} = \frac{5.584\,VN}{W}$$

where

V = ml. of potassium permanganate solution used.
N = normality of potassium permanganate solution.
W = weight of sample.

F–4c. *Aluminum.*—Calculate the percentage of aluminum in the sample as follows:

$$\text{Percentage of aluminum} = \frac{(A - 0.0143BW)\,52.91}{W}$$

where

A = weight of the combined precipitate. (See par. F–4b.)
B = percent iron. (See par. F–4b.)
W = weight of sample.

F–4d. *Granulation.*—

F–4d (1). *Method for other than Class C.*—Place an accurately weighed portion of approximately 100 gm. of the sample on the specified nest of sieves properly superimposed and assembled with a bottom pan. Cover and shake for 5 minutes by hand, or mechanically by means of a mechanical shaker geared to produce 300±15 gyrations and 150±10 taps of the striker per minute. Weigh the portions retained or passed by the various sieves and calculate the results to a percentage basis as required.

F–4d (2). *Method for Class C only.*—Weigh approximately 50.0 gm. of the sample and transfer to a 62-micron (No. 230) U. S. standard sieve. Wash the material with a stream of water until the water comes through clear. Dry the residue on the screen for 20 to 30 minutes in an oven at 105° to 120° C. Transfer the dried residue to an 88-micron (No. 170) U. S. standard sieve which is nested above a 62-micron (No. 230) U. S. standard sieve and a bottom pan. Cover and shake for 15 minutes by means of a mechanical shaker geared to produce 300±15 gyrations and 150±10 taps of the striker per minute. Weigh the material retained on each sieve and calculate the results to a percentage basis.

F–5. *Retests.*—If the official sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having analysis of each primary sample made at his own expense. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and submit the lot for acceptance, provided the markings on the container are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector.

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G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—

G–1a. *For domestic shipment.*—Unless otherwise specified powdered silicon shall be packed in standard commercial containers so constructed as to insure safe transportation and warehousing.

G–1b. *For overseas shipment.*—Unless otherwise specified, powdered silicon shall be packed in multiwall paper shipping sacks, 55-gallon removable head, steel drums, 25-gallon steel drums, or in textile laminated bags. Multiwall paper shipping sacks shall conform to the requirements of Joint Army-Navy Specification JAN–P–118, 55-gallon steel drums shall conform to the requirements of Joint Army-Navy Specification JAN–P–110, and 25-gallon steel drums shall conform to the requirements of the Interstate Commerce Commission Specification 37E. Textile laminated shipping bags shall be in conformance with best commercial practice and shall be constructed of 10-ounce weight burlap, 40 inches wide; or osnaburg of 2.35 yards per pound weight, 36 inches wide. The ply construction from the inside to the outside shall be paper-asphalt-paper-asphalt-textile. The gross weight of powdered silicon in textile laminated bags or multiwall paper shipping sacks shall not exceed 75 pounds.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisitions, schedules and contracts or orders should contain the following information:

(a) Title of the specification, the number, and date.

(b) Grade and class of powdered silicon required. (See par. B–1.)

H–2. *Uses.*—Silicon covered by this specification is intended for use as an ingredient in pyrotechnic composition.

H–3. Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–4. Copies of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles by Freight, of which I. C. C. Specification 37E forms a part, may be obtained upon application accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 40 cents.

H–5. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases) and Navy Department speci-

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fications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of this specification required for Army purchases may be obtained from the office of Chief of Ordnance, Army Service Forces, Washington 25, D. C.

H–8. Copies of this specification required for Navy purchases may be obtained as indicated in paragraph H–6.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

NAVY:OS.

[JAN–S–230]

(no. 105)

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